OGRDBstats Report

Contents

1 Novel sequence analysis **2**
1.1 End-nucleotide composition 2
1.2 Per-nucleotide consensus where previous nucleotides match the consensus 4
1.3 Whole-sequence composition of each assigned read 6
1.4 Final three nucleotides: frequency of each observed triplet 7
1.5 CDR3 length distribution, in assignments to novel alleles 8

2 Variation from germline, in assignments to each allele **10**

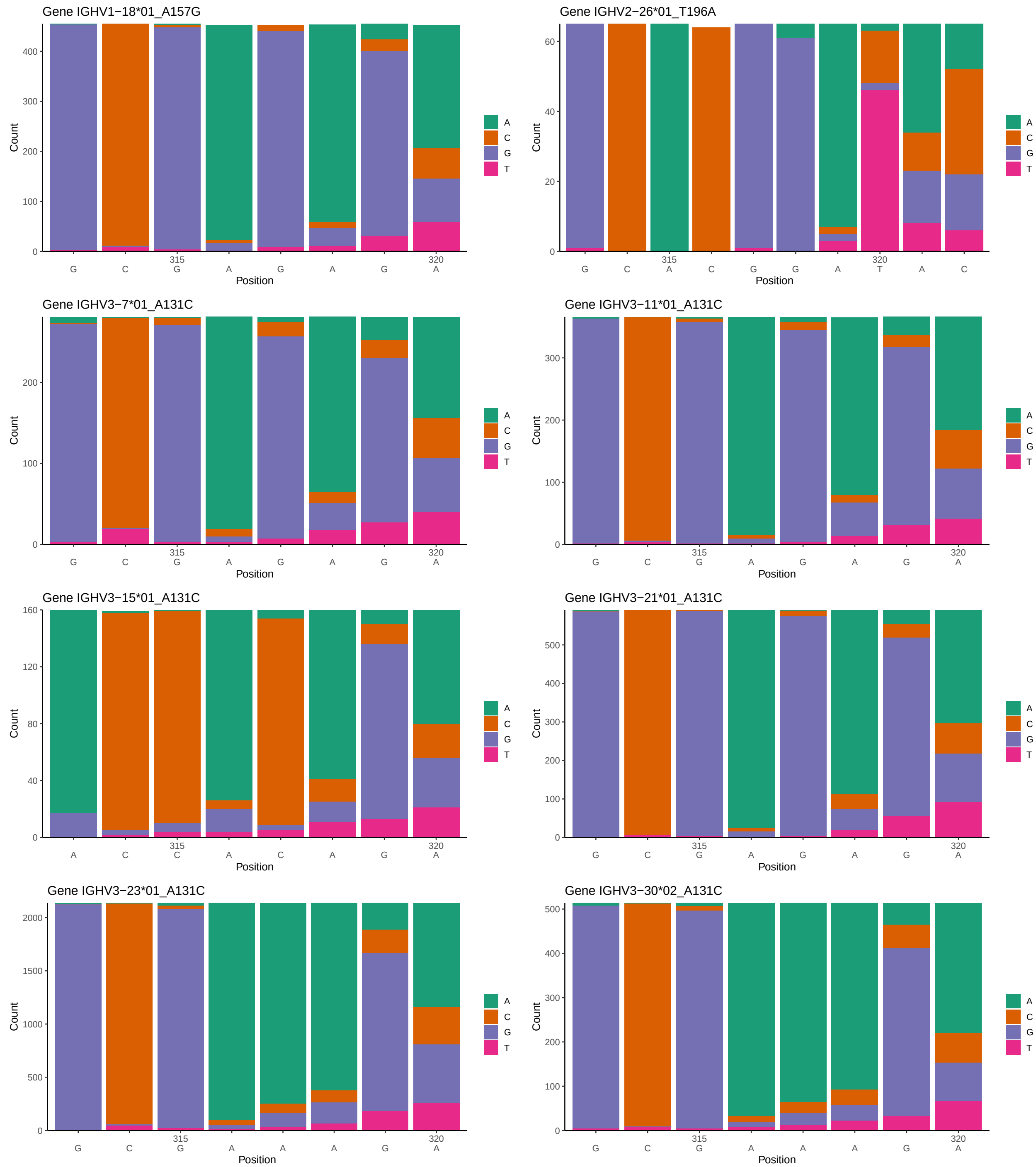
3 Allele usage in potential haplotype anchor genes **16**

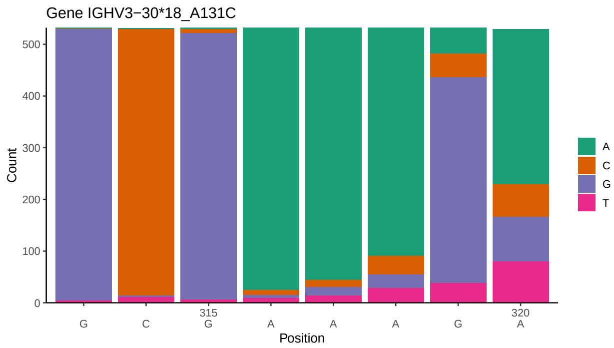
4 Haplotype plots **17**

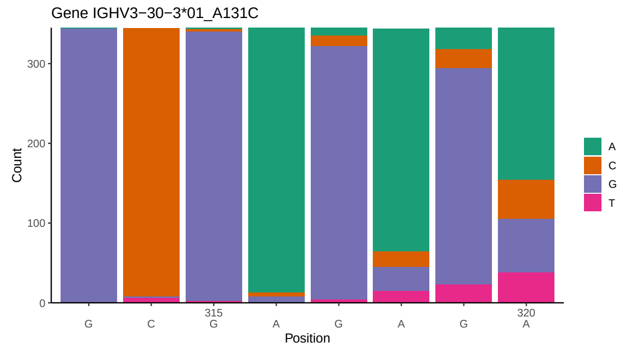
5 Configuration settings **18**

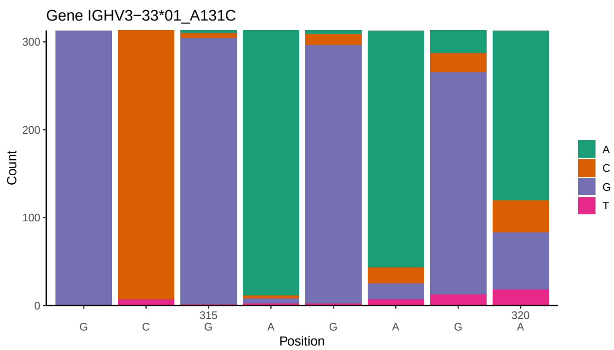
1 Novel sequence analysis

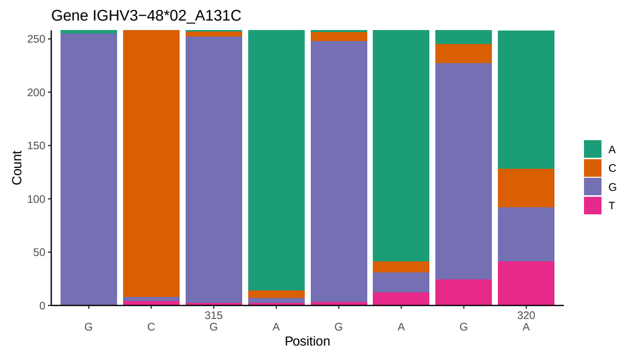
1.1 End-nucleotide composition

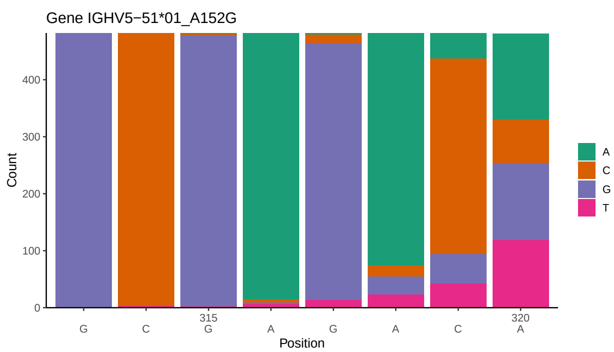



Gene IGHV3−30*18_A131C
Count
0
100
200
300
400
500
315
320
G C G A A A G A
Position
A
C
G
T

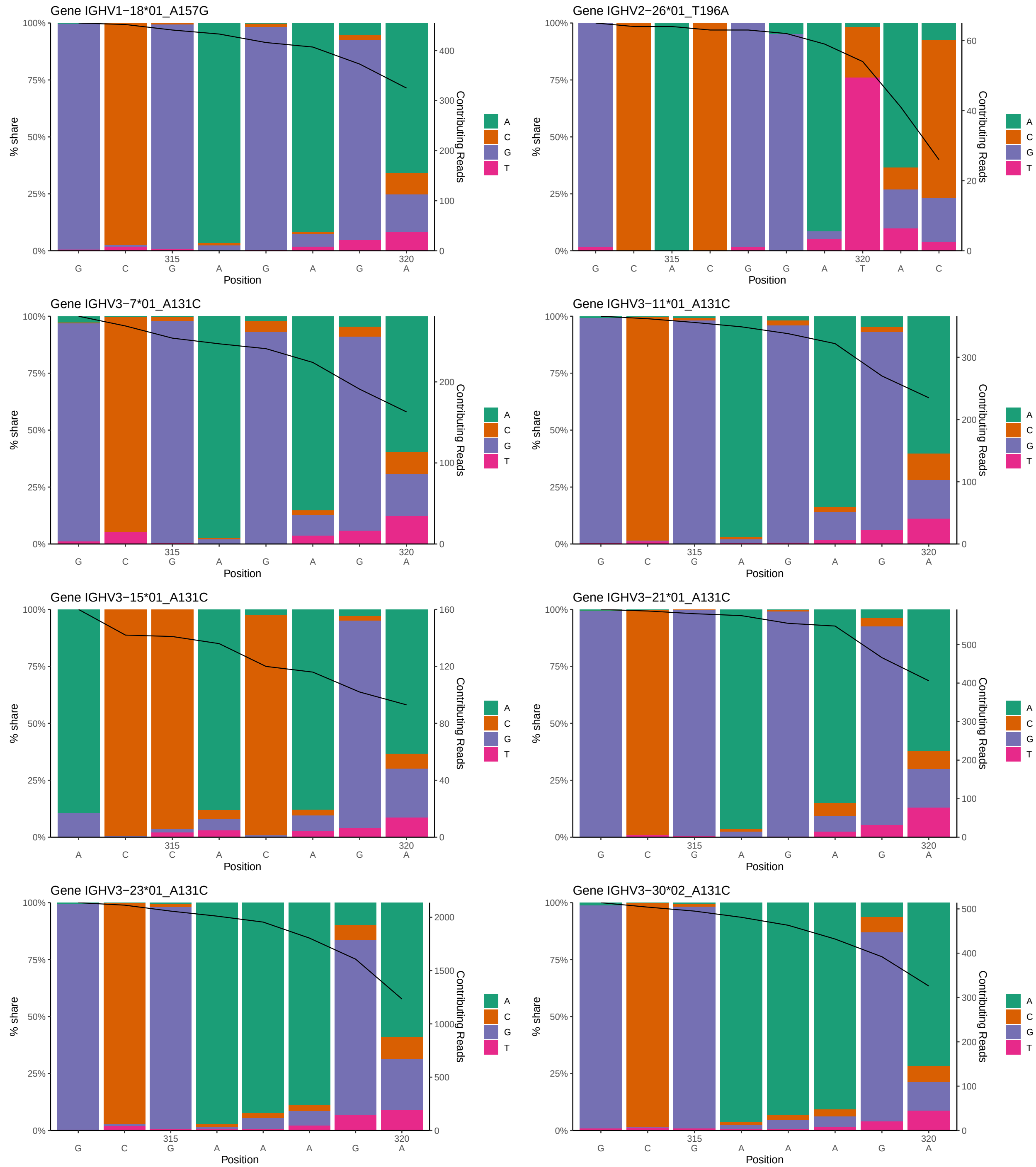

Gene IGHV3−30−3*01_A131C
Count
0
100
200
300
315
320
G C G A G A G A
Position
A
C
G
T

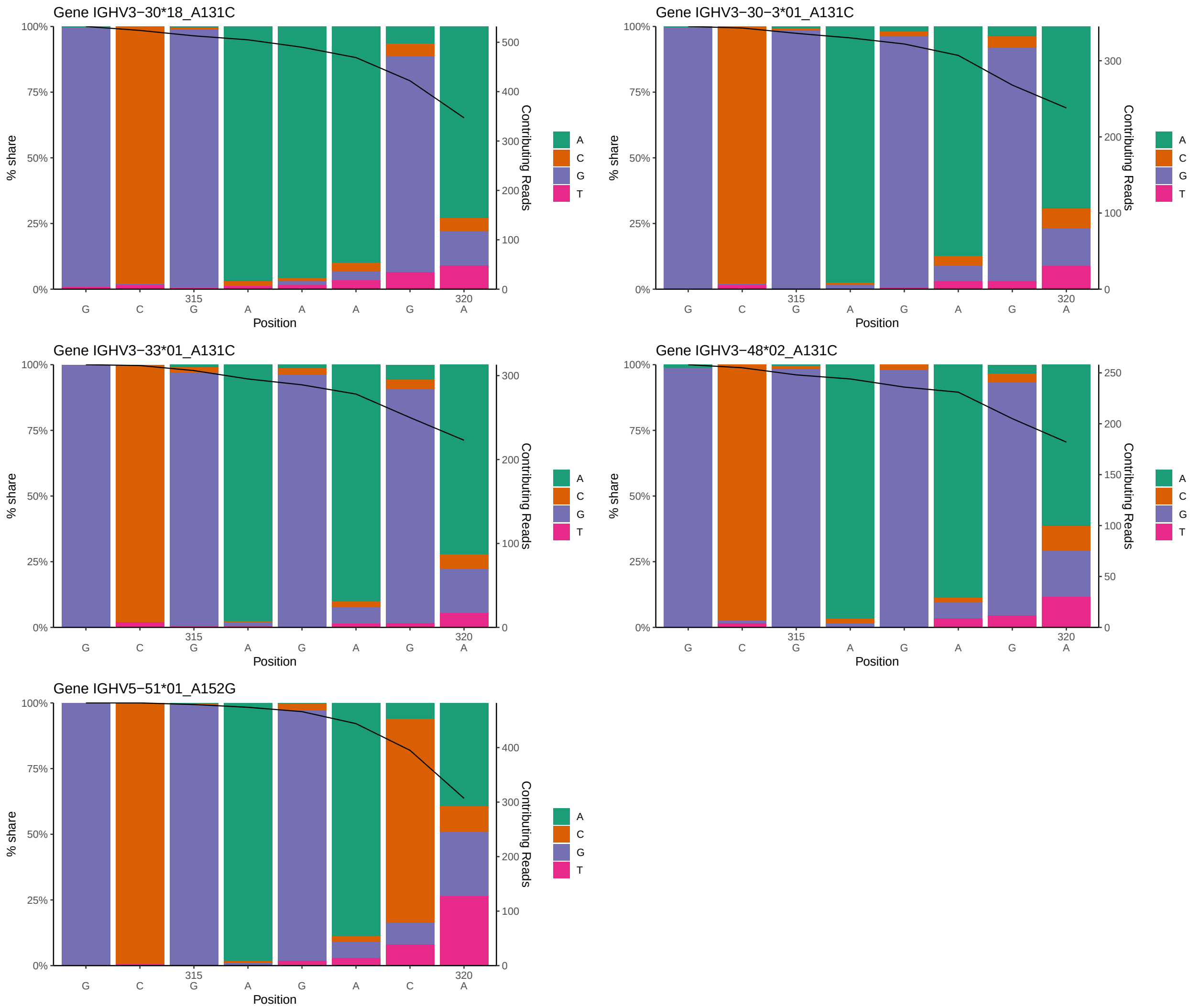

Gene IGHV3−33*01_A131C
Count
0
100
200
300
315
320
G C G A G A G A
Position
A
C
G
T


Gene IGHV3−48*02_A131C
Count
0
50
100
150
200
250
315
320
G C G A G A G A
Position
A
C
G
T

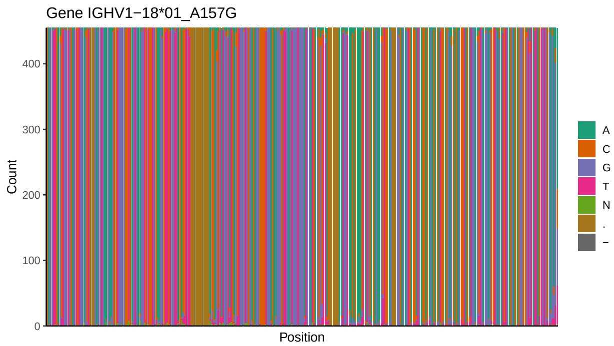

Gene IGHV5−51*01_A152G
Count
0
100
200
300
400
315
320
G C G A G A C A
Position
A
C
G
T

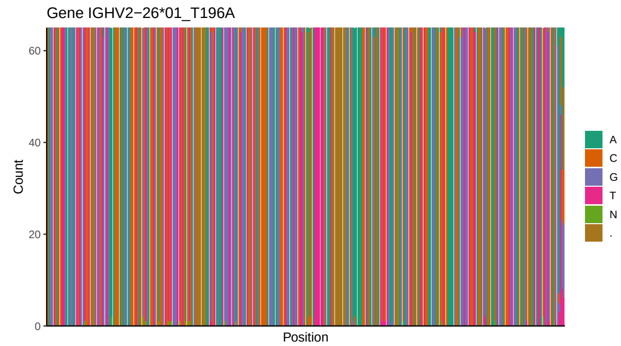
1.2 Per-nucleotide consensus where previous nucleotides match the consensus

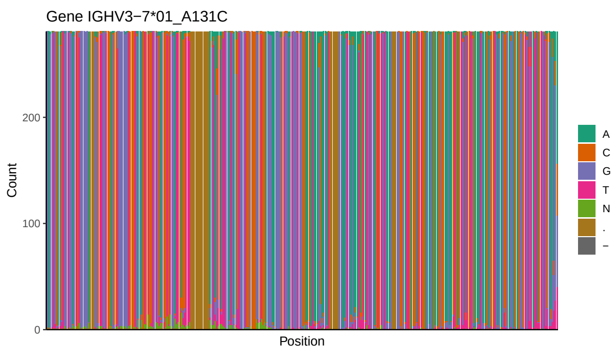


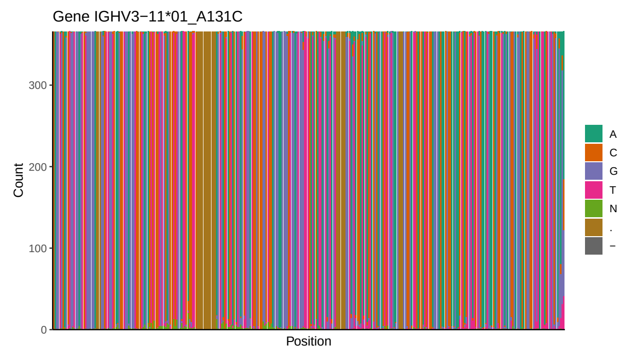

Gene IGHV3−30*18_A131C
Gene IGHV3−30−3*01_A131C
Gene IGHV3−33*01_A131C
Gene IGHV3−48*02_A131C
Gene IGHV5−51*01_A152G
% share
Contributing Reads
Position
100%
75%
50%
25%
0%
315
320
A
C
G
T

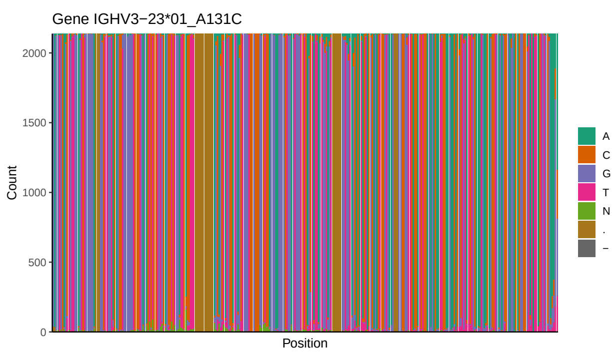
1.3 Whole-sequence composition of each assigned read

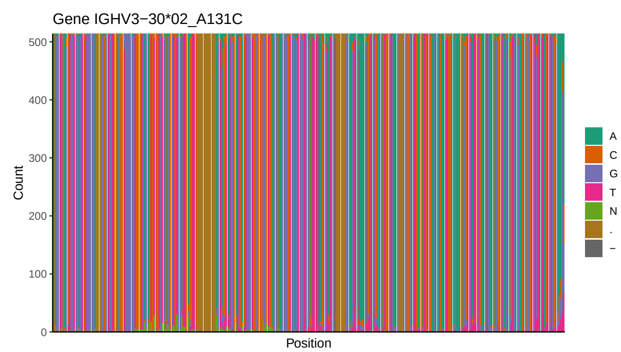


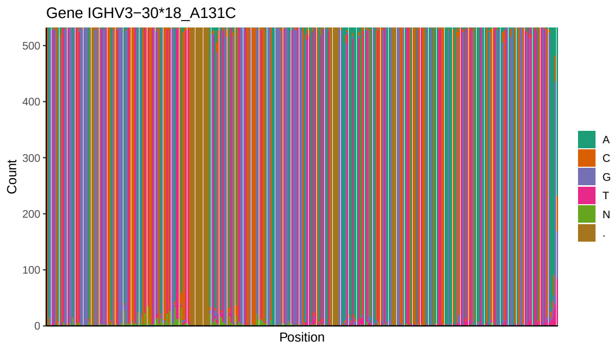


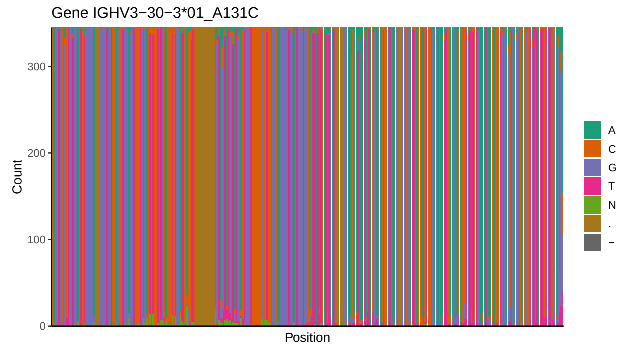


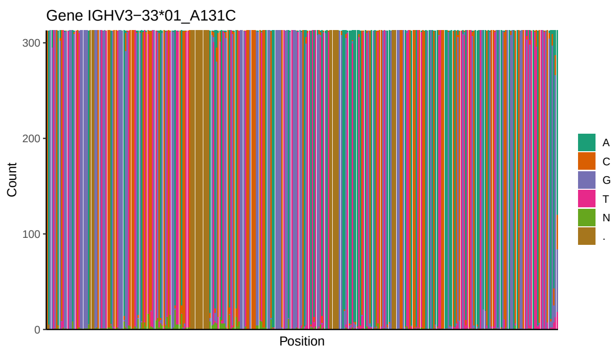


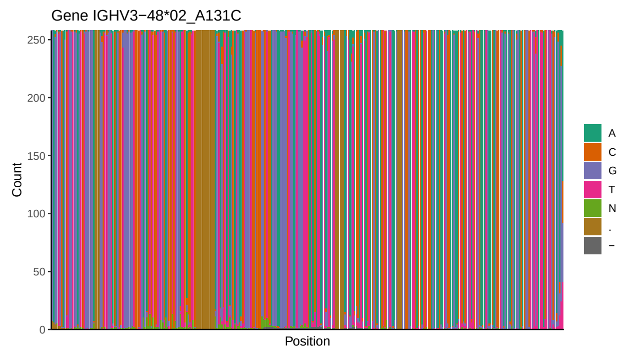


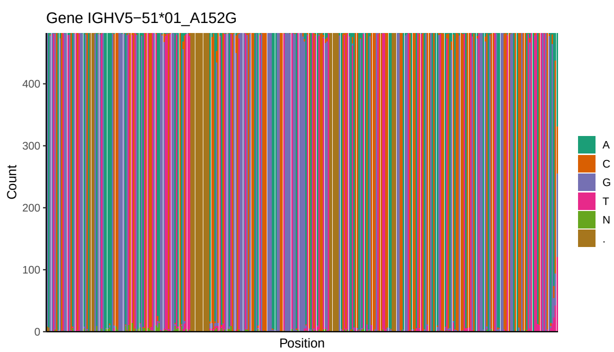




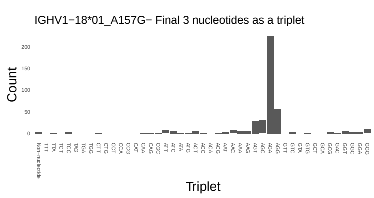


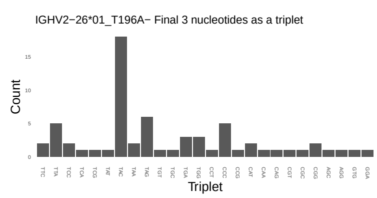


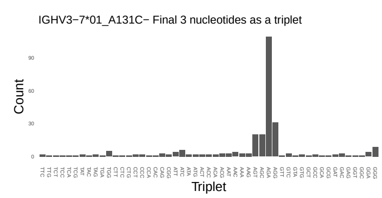


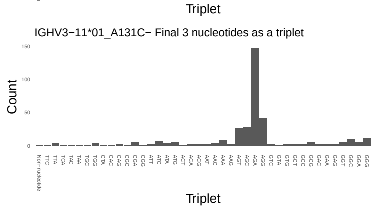


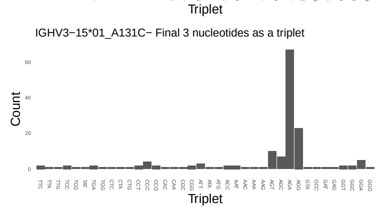
1.4 Final three nucleotides: frequency of each observed triplet

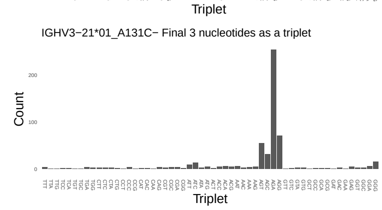


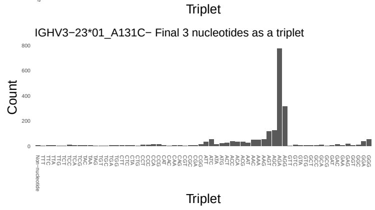


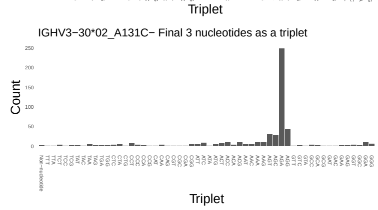


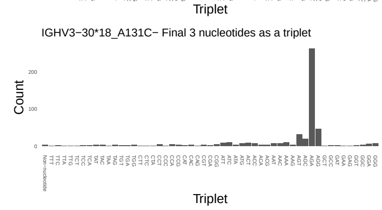


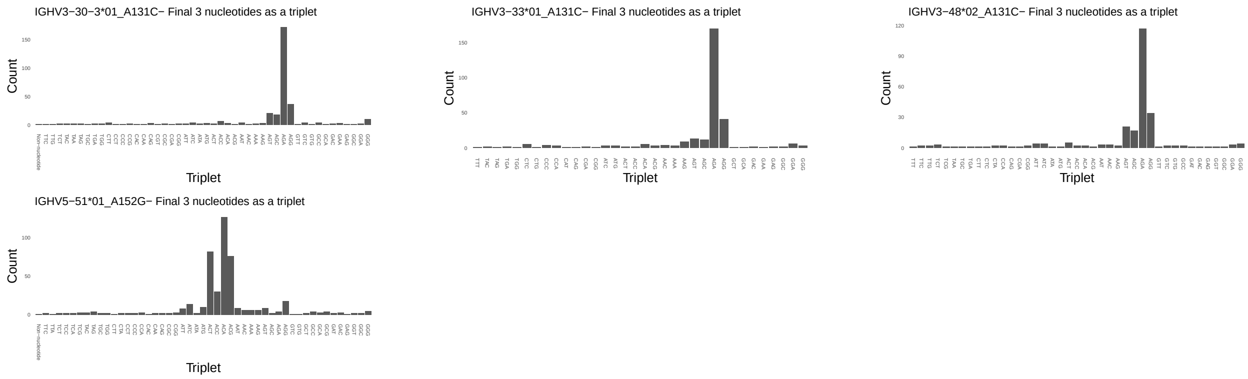




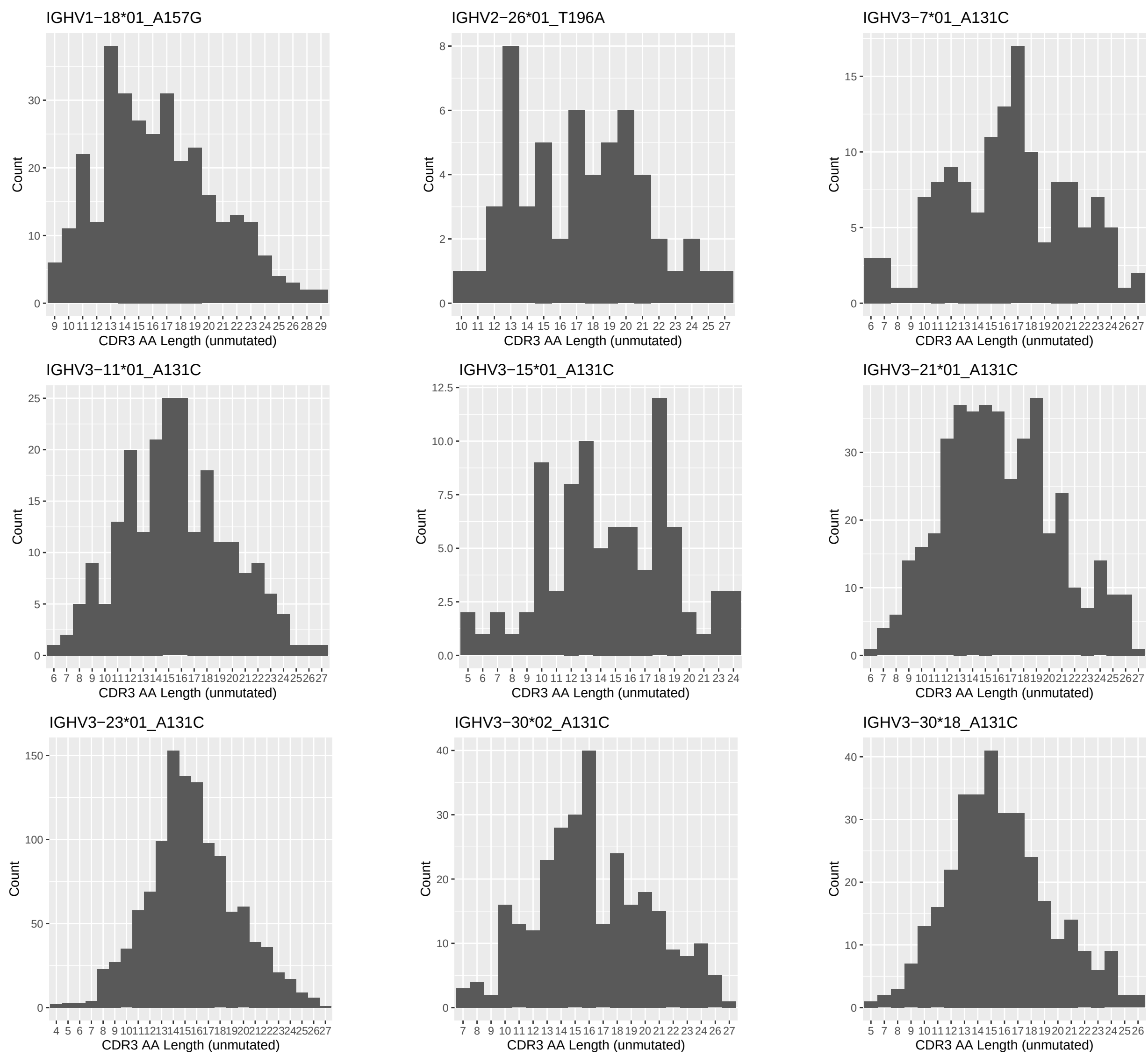


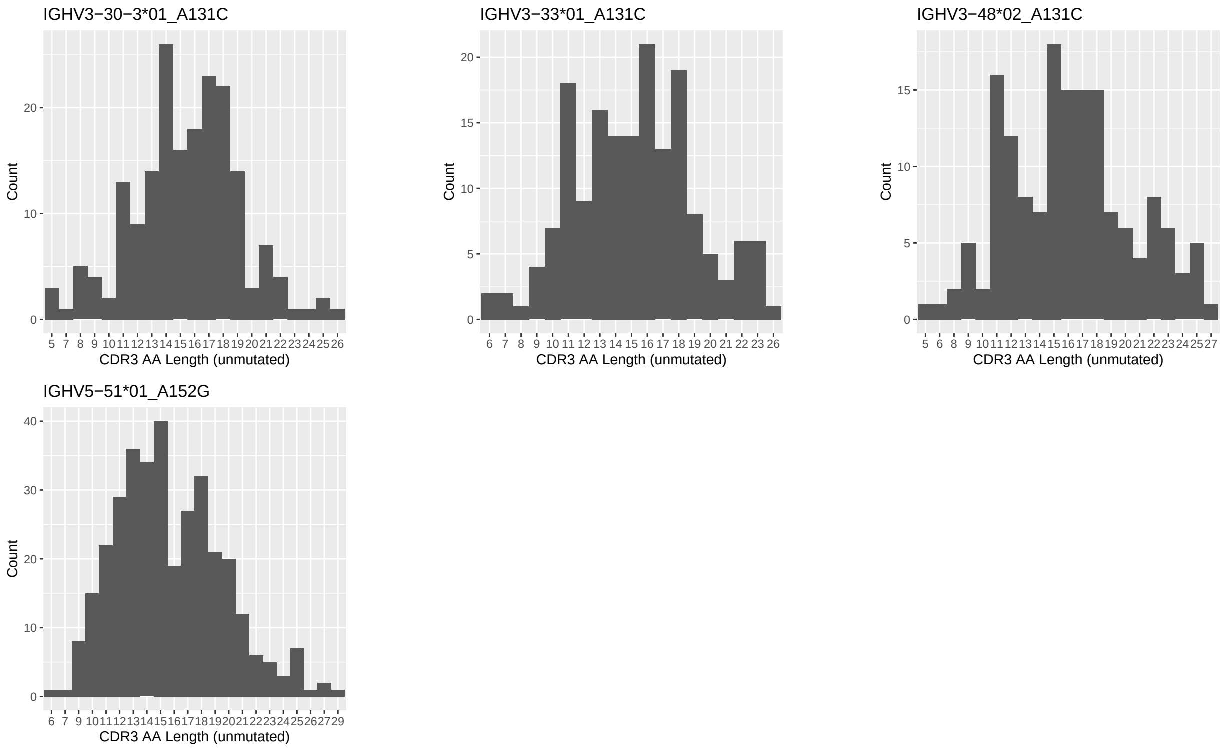




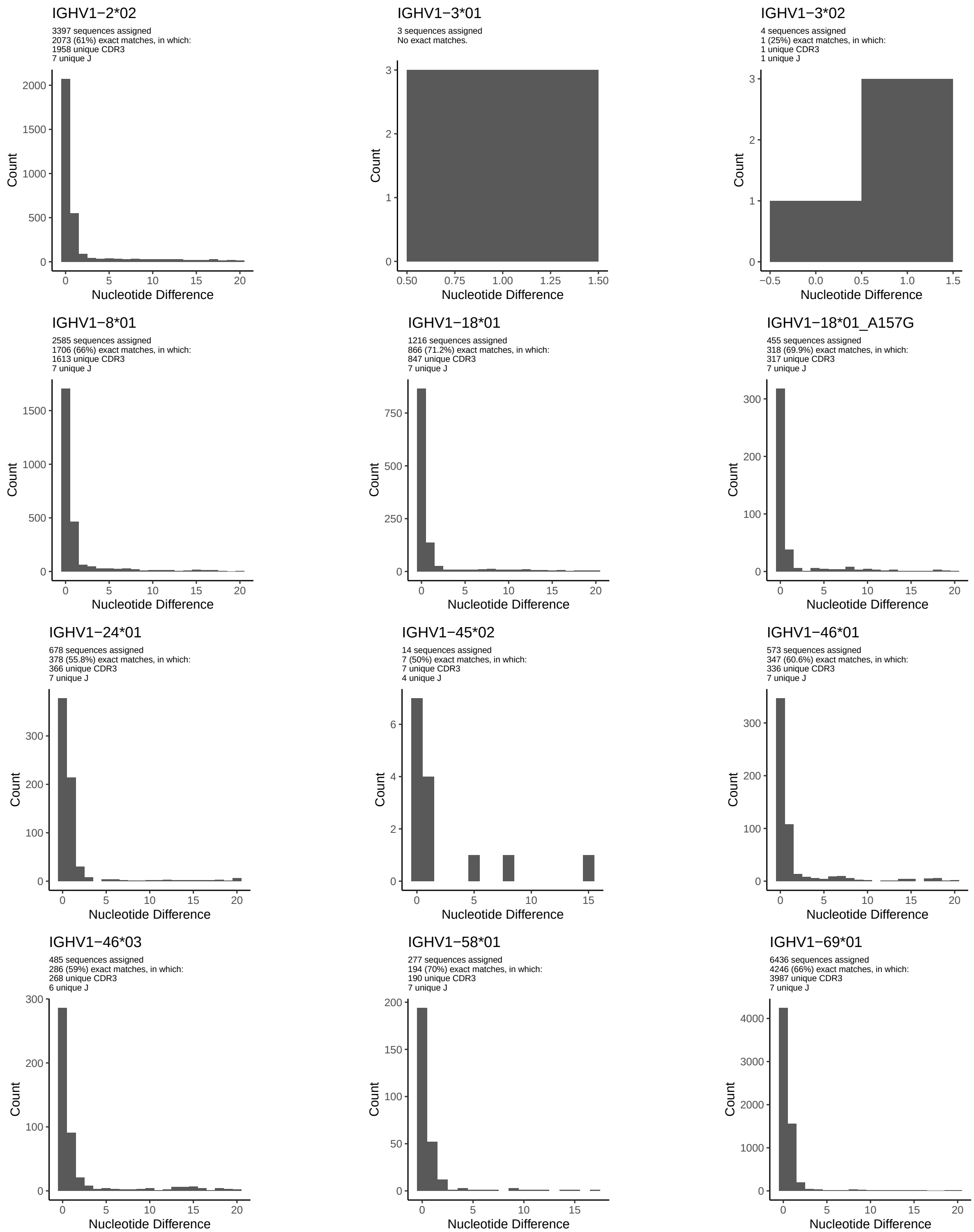


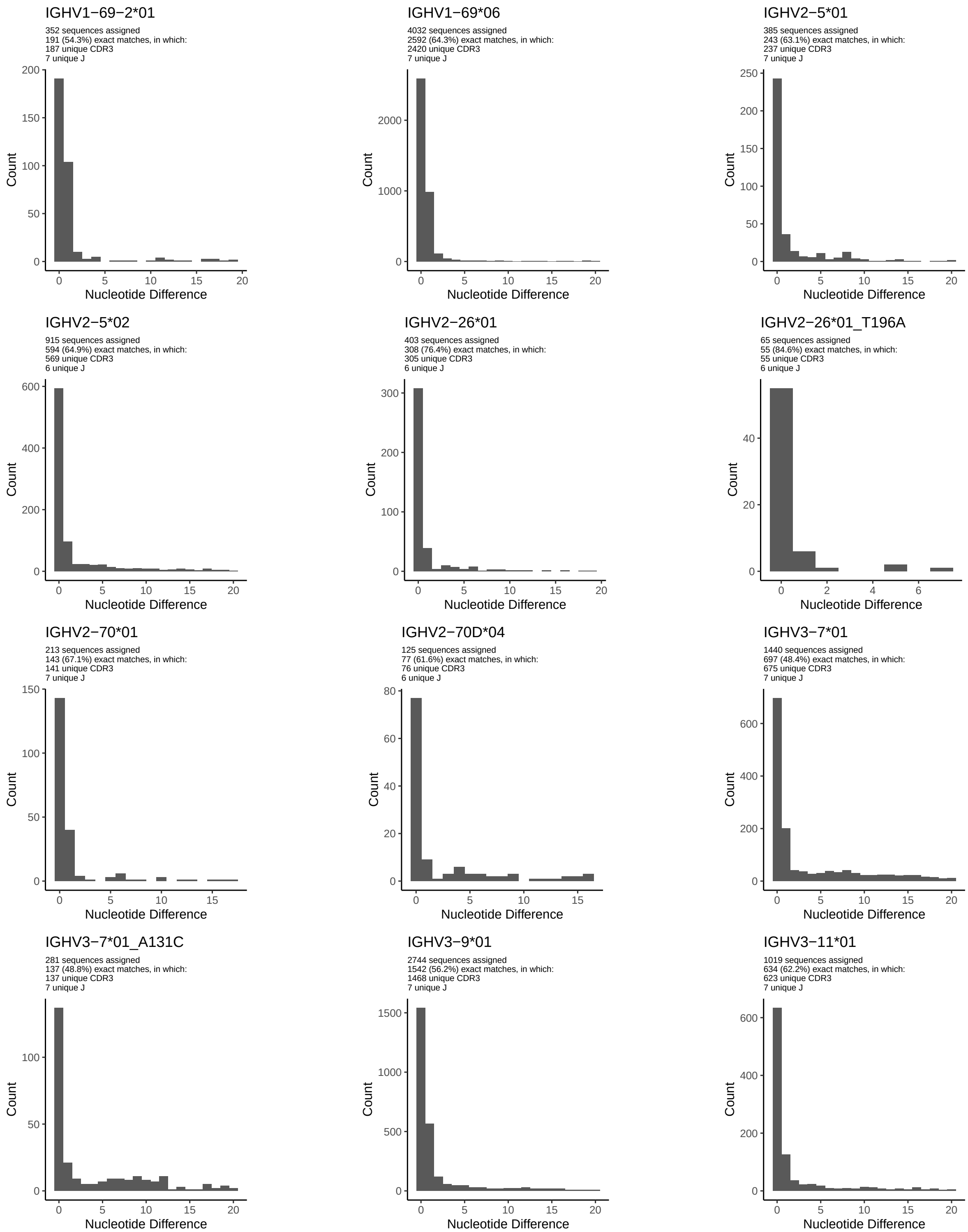
1.5 CDR3 length distribution, in assignments to novel alleles

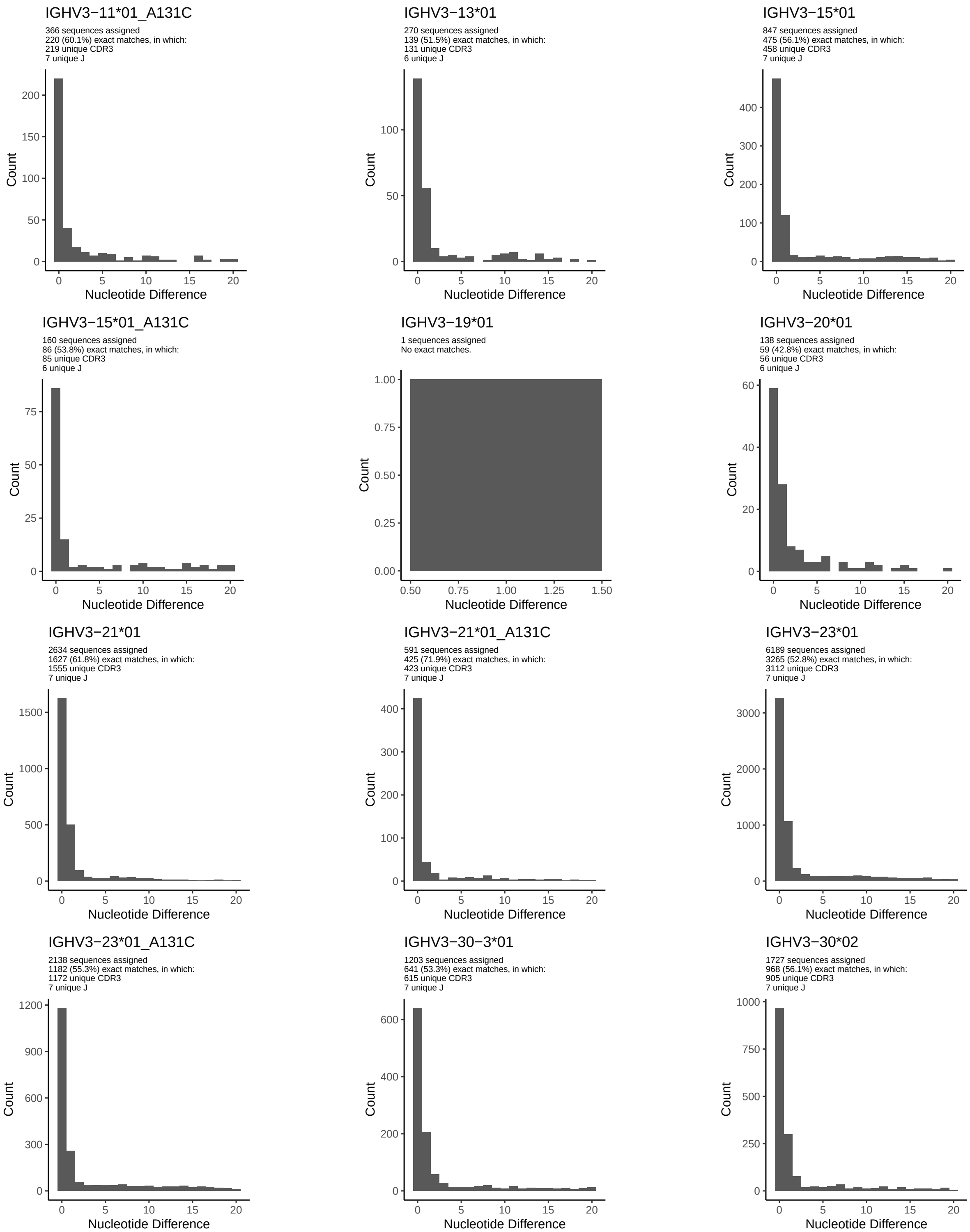


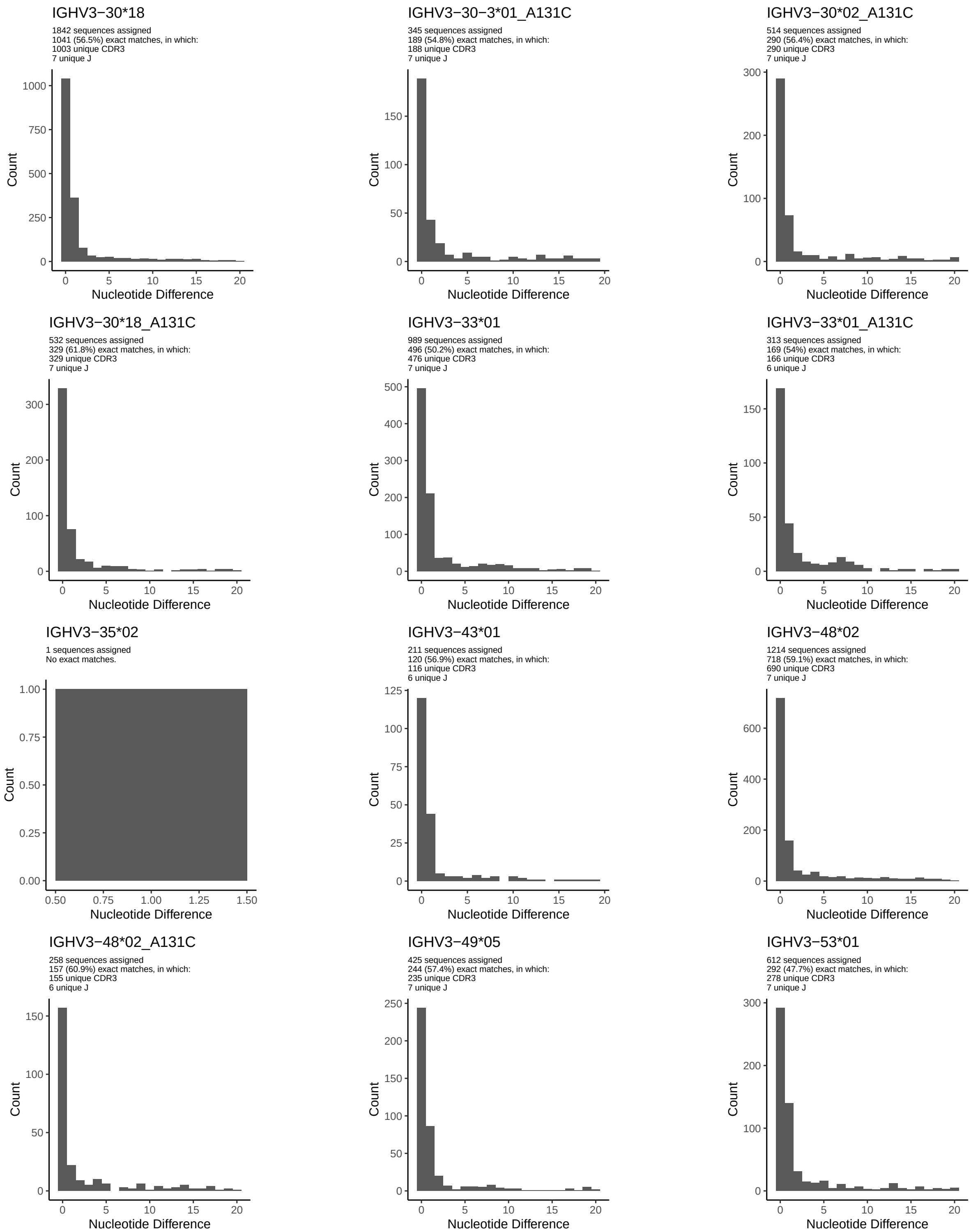

IGHV3−30−3*01_A131C
IGHV3−33*01_A131C
IGHV3−48*02_A131C
IGHV5−51*01_A152G
Count
CDR3 AA Length (unmutated)

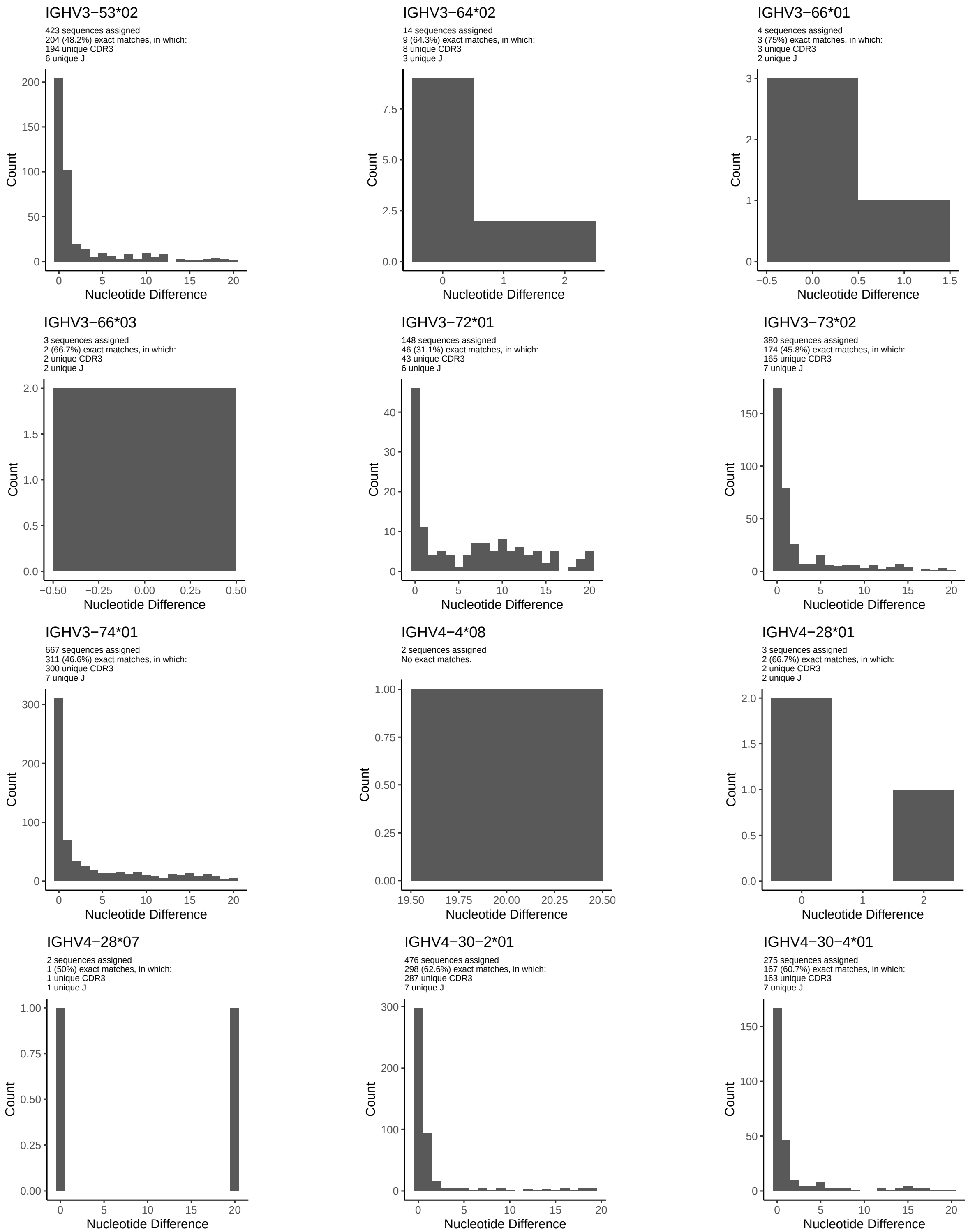
2 Variation from germline, in assignments to each allele

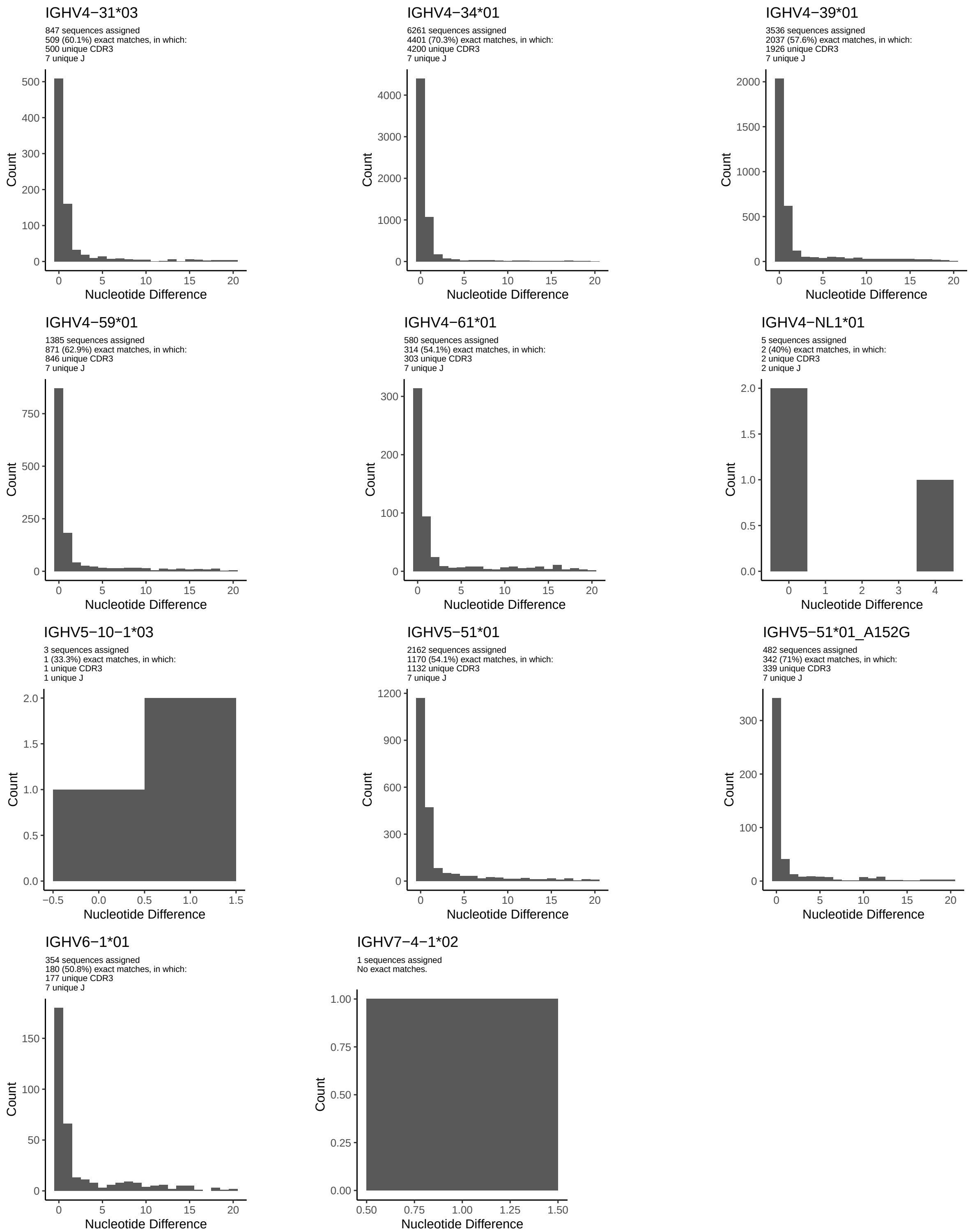



IGHV1−69−2*01
352 sequences assigned
191 (54.3%) exact matches, in which:
187 unique CDR3
7 unique J
IGHV1−69*06
4032 sequences assigned
2592 (64.3%) exact matches, in which:
2420 unique CDR3
7 unique J
IGHV2−5*01
385 sequences assigned
243 (63.1%) exact matches, in which:
237 unique CDR3
7 unique J
IGHV2−5*02
915 sequences assigned
594 (64.9%) exact matches, in which:
569 unique CDR3
6 unique J
IGHV2−26*01
403 sequences assigned
308 (76.4%) exact matches, in which:
305 unique CDR3
6 unique J
IGHV2−26*01_T196A
65 sequences assigned
55 (84.6%) exact matches, in which:
55 unique CDR3
6 unique J
IGHV2−70*01
213 sequences assigned
143 (67.1%) exact matches, in which:
141 unique CDR3
7 unique J
IGHV2−70D*04
125 sequences assigned
77 (61.6%) exact matches, in which:
76 unique CDR3
6 unique J
IGHV3−7*01
1440 sequences assigned
697 (48.4%) exact matches, in which:
675 unique CDR3
7 unique J
IGHV3−7*01_A131C
281 sequences assigned
137 (48.8%) exact matches, in which:
137 unique CDR3
7 unique J
IGHV3−9*01
2744 sequences assigned
1542 (56.2%) exact matches, in which:
1468 unique CDR3
7 unique J
IGHV3−11*01
1019 sequences assigned
634 (62.2%) exact matches, in which:
623 unique CDR3
7 unique J
Count
Nucleotide Difference

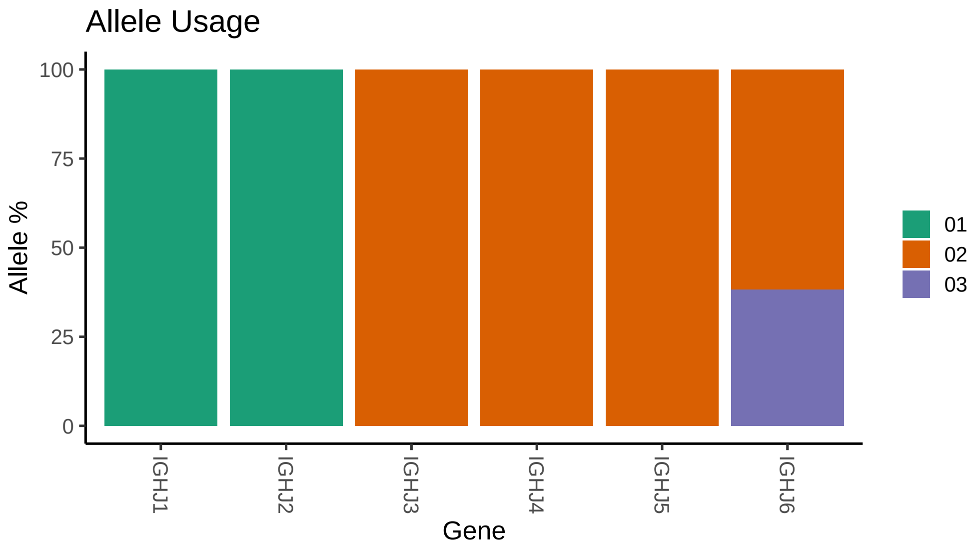

IGHV3−11*01_A131C
366 sequences assigned
220 (60.1%) exact matches, in which:
219 unique CDR3
7 unique J
IGHV3−13*01
270 sequences assigned
139 (51.5%) exact matches, in which:
131 unique CDR3
6 unique J
IGHV3−15*01
847 sequences assigned
475 (56.1%) exact matches, in which:
458 unique CDR3
7 unique J
IGHV3−15*01_A131C
160 sequences assigned
86 (53.8%) exact matches, in which:
85 unique CDR3
6 unique J
IGHV3−19*01
1 sequences assigned
No exact matches.
IGHV3−20*01
138 sequences assigned
59 (42.8%) exact matches, in which:
56 unique CDR3
6 unique J
IGHV3−21*01
2634 sequences assigned
1627 (61.8%) exact matches, in which:
1555 unique CDR3
7 unique J
IGHV3−21*01_A131C
591 sequences assigned
425 (71.9%) exact matches, in which:
423 unique CDR3
7 unique J
IGHV3−23*01
6189 sequences assigned
3265 (52.8%) exact matches, in which:
3112 unique CDR3
7 unique J
IGHV3−23*01_A131C
2138 sequences assigned
1182 (55.3%) exact matches, in which:
1172 unique CDR3
7 unique J
IGHV3−30−3*01
1203 sequences assigned
641 (53.3%) exact matches, in which:
615 unique CDR3
7 unique J
IGHV3−30*02
1727 sequences assigned
968 (56.1%) exact matches, in which:
905 unique CDR3
7 unique J
Count
Nucleotide Difference


IGHV3−30*18
1842 sequences assigned
1041 (56.5%) exact matches, in which:
1003 unique CDR3
7 unique J
IGHV3−30−3*01_A131C
345 sequences assigned
189 (54.8%) exact matches, in which:
188 unique CDR3
7 unique J
IGHV3−30*02_A131C
514 sequences assigned
290 (56.4%) exact matches, in which:
290 unique CDR3
7 unique J
IGHV3−30*18_A131C
532 sequences assigned
329 (61.8%) exact matches, in which:
329 unique CDR3
7 unique J
IGHV3−33*01
989 sequences assigned
496 (50.2%) exact matches, in which:
476 unique CDR3
7 unique J
IGHV3−33*01_A131C
313 sequences assigned
169 (54%) exact matches, in which:
166 unique CDR3
6 unique J
IGHV3−35*02
1 sequences assigned
No exact matches.
IGHV3−43*01
211 sequences assigned
120 (56.9%) exact matches, in which:
116 unique CDR3
6 unique J
IGHV3−48*02
1214 sequences assigned
718 (59.1%) exact matches, in which:
690 unique CDR3
7 unique J
IGHV3−48*02_A131C
258 sequences assigned
157 (60.9%) exact matches, in which:
155 unique CDR3
6 unique J
IGHV3−49*05
425 sequences assigned
244 (57.4%) exact matches, in which:
235 unique CDR3
7 unique J
IGHV3−53*01
612 sequences assigned
292 (47.7%) exact matches, in which:
278 unique CDR3
7 unique J
Count
Nucleotide Difference

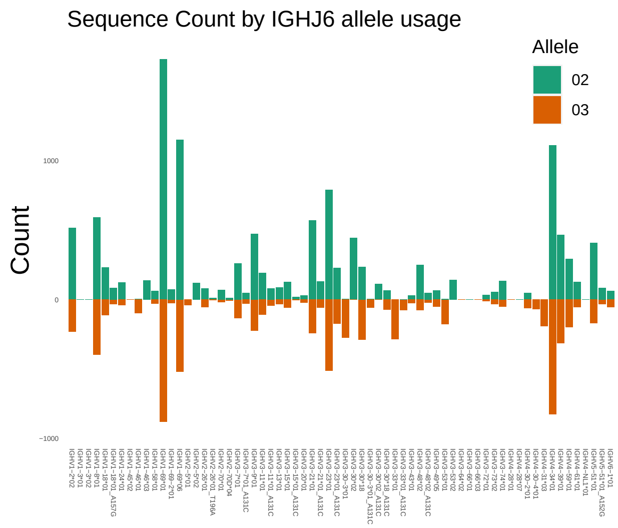

IGHV3−53*02
423 sequences assigned
204 (48.2%) exact matches, in which:
194 unique CDR3
6 unique J
IGHV3−64*02
14 sequences assigned
9 (64.3%) exact matches, in which:
8 unique CDR3
3 unique J
IGHV3−66*01
4 sequences assigned
3 (75%) exact matches, in which:
3 unique CDR3
2 unique J
IGHV3−66*03
3 sequences assigned
2 (66.7%) exact matches, in which:
2 unique CDR3
2 unique J
IGHV3−72*01
148 sequences assigned
46 (31.1%) exact matches, in which:
43 unique CDR3
6 unique J
IGHV3−73*02
380 sequences assigned
174 (45.8%) exact matches, in which:
165 unique CDR3
7 unique J
IGHV3−74*01
667 sequences assigned
311 (46.6%) exact matches, in which:
300 unique CDR3
7 unique J
IGHV4−4*08
2 sequences assigned
No exact matches.
IGHV4−28*01
3 sequences assigned
2 (66.7%) exact matches, in which:
2 unique CDR3
2 unique J
IGHV4−28*07
2 sequences assigned
1 (50%) exact matches, in which:
1 unique CDR3
1 unique J
IGHV4−30−2*01
476 sequences assigned
298 (62.6%) exact matches, in which:
287 unique CDR3
7 unique J
IGHV4−30−4*01
275 sequences assigned
167 (60.7%) exact matches, in which:
163 unique CDR3
7 unique J
Count
Nucleotide Difference


IGHV4−31*03
847 sequences assigned
509 (60.1%) exact matches, in which:
500 unique CDR3
7 unique J
IGHV4−34*01
6261 sequences assigned
4401 (70.3%) exact matches, in which:
4200 unique CDR3
7 unique J
IGHV4−39*01
3536 sequences assigned
2037 (57.6%) exact matches, in which:
1926 unique CDR3
7 unique J
IGHV4−59*01
1385 sequences assigned
871 (62.9%) exact matches, in which:
846 unique CDR3
7 unique J
IGHV4−61*01
580 sequences assigned
314 (54.1%) exact matches, in which:
303 unique CDR3
7 unique J
IGHV4−NL1*01
5 sequences assigned
2 (40%) exact matches, in which:
2 unique CDR3
2 unique J
IGHV5−10−1*03
3 sequences assigned
1 (33.3%) exact matches, in which:
1 unique CDR3
1 unique J
IGHV5−51*01
2162 sequences assigned
1170 (54.1%) exact matches, in which:
1132 unique CDR3
7 unique J
IGHV5−51*01_A152G
482 sequences assigned
342 (71%) exact matches, in which:
339 unique CDR3
7 unique J
IGHV6−1*01
354 sequences assigned
180 (50.8%) exact matches, in which:
177 unique CDR3
7 unique J
IGHV7−4−1*02
1 sequences assigned
No exact matches.
Count
Nucleotide Difference

3 Allele usage in potential haplotype anchor genes



4 Haplotype plots



5 Configuration settings

```
## Repertoire file: /misc/work/jenkins/PRJNA338795/AR04/AR04/AR04/AR04/ee/5a20e04dd2af0b1f988c992edff8a9
##
## Germline reference file: /misc/work/jenkins/PRJNA338795/AR04/AR04/AR04/AR04/ee/5a20e04dd2af0b1f988c99
##
## Novel allele file: /misc/work/jenkins/PRJNA338795/AR04/AR04/AR04/AR04/ee/5a20e04dd2af0b1f988c992edff8
##
## Species: Homosapiens
##
## Chain: IGHV
##
## Segment: V
```